

Deploying on Vercel

System: Jima Aba Jifar — Cafe & Restaurant Management System
Document: Alternative Deployment Guide v1.1 · **Updated:** August 19, 2026

1. What goes where

Status: BLOCKED. This is an alternative deployment reference, not approval to deploy. Complete all mandatory controls in documents 10-12 before using these instructions. Current blockers include schema synchronization, authentication defaults, startup account mutation, wildcard CORS, and incomplete regression evidence.

Vercel is built for frontend frameworks. This system has **two services**, so the recommended split is:

Piece	Where it runs
Next.js frontend (artifacts/nextjs-app)	Vercel — first-class Next.js support.
NestJS backend (artifacts/nestjs-server)	A long-running Node host: Railway, Render, Fly.io , or a small VM. (Vercel serverless functions are not a good fit for a full NestJS app with TypeORM connections.)
PostgreSQL	Managed database: Neon , Supabase, Railway Postgres, or RDS.

2. Step 1 — Database

1. Create a PostgreSQL database (e.g. Neon free tier).

2. Copy the connection string, e.g.

```
postgres://user:pass@host/db?sslmode=require — keep  
sslmode=require ; the backend enables SSL automatically when it  
sees it.
```

3. Step 2 — Backend (Railway example)

1. Push the repository to GitHub.
2. In Railway: **New Project → Deploy from GitHub**, root directory `artifacts/nestjs-server`.
3. Build & start commands:

```
Build:  pnpm install && pnpm run build  
Start:  node dist/main.js
```

4. Environment variables:

```
DATABASE_URL = postgres://...?sslmode=require  
JWT_SECRET   = <long random string>  
PORT          = 3001    (or leave to platform default)
```

5. Deploy only after the release gate is approved. Verify the protected health and API behavior without exposing Swagger publicly.
6. Provision initial production accounts through the approved controlled process. Do not call the development seed endpoint against production.

4. Step 3 — Frontend on Vercel

1. In Vercel: **Add New → Project**, import the GitHub repository.
2. **Root Directory:** set to `artifacts/nextjs-app` (Vercel detects Next.js automatically).
3. **Install command:** `pnpm install` (Vercel supports pnpm monorepos; enable "Include files outside root directory" if prompted).
4. **Environment variables:**

```
BACKEND_URL = https://habesha-api.up.railway.app
```

The frontend's `next.config.js` rewrite forwards `/backend/api/*` to this URL. Backend CORS must still be restricted as defense in depth and for any direct operational access.

5. Click **Deploy**. Your app is live at `https://<project>.vercel.app`.

5. Step 4 — Verify

Check	Expected
Open the Vercel URL	Login page with photo background loads.
Log in as admin	Dashboard renders with data (KPI cards populated).
Browser dev tools → Network	API calls go to <code>/backend/api/...</code> on the same domain and return 200.
Create a test order	Appears on the Order Progress Board within 10 s.

6. Custom domain (optional)

1. Vercel → Project → Settings → Domains → add `app.yourdomain.com` and follow the DNS instructions.
2. HTTPS certificates are automatic.

7. Updates

Every push to the connected GitHub branch redeploys the frontend automatically on Vercel (and the backend on Railway). Preview deployments are created for pull requests.

Before going live: complete the production checklist and evidence matrix in documents 05 and 12. Do not deploy while any mandatory item is marked BLOCKED.

8. Troubleshooting

Symptom	Likely cause / fix
Login fails with network error	<code>BACKEND_URL</code> missing or wrong on Vercel; redeploy after fixing (rewrites are baked at build time).
500s from API	Backend can't reach the database — check <code>DATABASE_URL</code> and that <code>sslmode=require</code> is present for managed Postgres.
Build fails on Vercel	Ensure Root Directory is <code>artifacts/nextjs-app</code> and <code>pnpm</code> is selected; clear build cache and retry.
Works locally, 404 on API in production	Calls must go through <code>/backend/api/*</code> ; hardcoded <code>localhost:3001</code> URLs won't work in production.